

Network Layer: Logical Addressing

IP Address Classes

- Class A: The first octet is the network portion. Octets 2, 3, and 4 are for subnets/hosts
- Class B: The first two octets are the network portion. Octets 3 and 4 are for subnets/hosts
- Class C: The first three octets are the network portion. Octet 4 is for subnets/hosts

Private Address Range

Address Class	Reserved Address Space
Class A	10.0.0.0 - 10.255.255.255
Class B	172.16.0.0 - 172.31.255.255
Class C	192.168.0.0 - 192.168.255.255

Network Masks

- Distinguishes which portion of the address identifies the network and which portion of the address identifies the node.
- Default masks:
 - Class A: 255.0.0.0
 - Class B: 255.255.0.0
 - Class C: 255.255.255.0

Subnetting

- Creates multiple logical networks that exist within a single Class A, B, or C network.
- If you do not subnet, you will only be able to use one network from your Class A, B, or C network, which is unrealistic

Benefits of Subnetting

- 1) Reduced network traffic
- 2) Optimized network performance
- 3) Simplified management

Practice Example #1: 255.255.255.128 (/25) Network 192.168.10.0

- How many subnets?

Since 128 is 1 bit on (10000000), the answer would be $2^1 = 2$.

ie., 192.168.10.0 & 192.168.10.128

- How many hosts per subnet?

We have 7 host bits off (10000000), so the equation would be $2^7 - 2 = 126$ hosts.

In 192.168.10.0 network → 192.168.10.0 – 192.168.10.127 (126 hosts)

In 192.168.10.128 network → 192.168.10.128 – 192.168.10.255 (126 hosts)

Practice Example #1: 255.255.255.128 (/25) Network 192.168.10.0

- What's the broadcast address for each subnet?

The number right before the value of the next subnet is all host bits turned on and equals the broadcast address. For the zero subnet, the next subnet is 128, so the broadcast of the 0 subnet is 127.

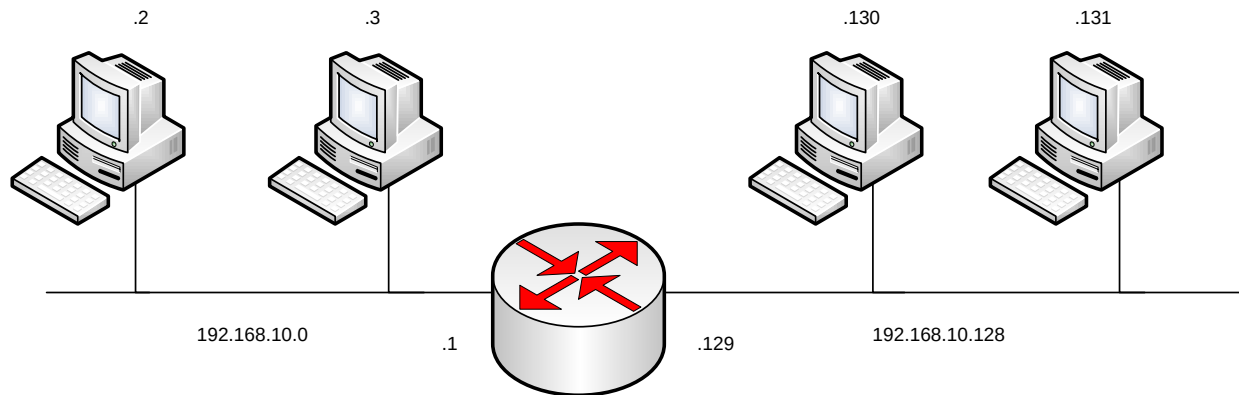
For 192.168.10.0 network → 192.168.10.127 is the broadcast address

For 192.168.10.128 network → 192.168.10.255 is the broadcast address

- What are the valid hosts?

These are the numbers between the subnet and broadcast address

Logical Network Implementation



Practice Example #2: 255.255.255.224 (/27) Network 192.168.10.0

- How many subnets?

224 is 11100000, so our equation would be $2^3 = 8$.

- How many hosts?

$2^5 - 2 = 30$.

- What are the valid subnets?

$256 - 224 = 32$. We just start at zero and count to the subnet mask value in blocks (increments) of 32:

0, 32, 64, 96, 128, 160, 192, and 224.

Practice Example #2: 255.255.255.224 (/27) Network 192.168.10.0

- What's the broadcast address for each subnet (always the number right before the next subnet)?
- What are the valid hosts (the numbers between the subnet number and the broadcast address)?

Practice Example #2: 255.255.255.224 (/27) Network 192.168.10.0

Subnet Address	0	32		192	224
First Host	1	33		193	225
Last Host	30	62		222	254
Broadcast Address	31	63		223	255

Practice Example #3: 255.255.128.0 (/17) Network 172.16.0.0

- Subnets? $2^1 = 2$
- Hosts? $2^{15} - 2 = 32,766$ (7 bits in the third octet, and 8 in the fourth)
- Valid subnets? $256 - 128 = 128$. 0, 128. Remember that subnetting is performed in the third octet, so the subnet numbers are 0.0 and 128.0
ie., 172.16.0.0 and 172.16.128.0
- Broadcast address for each subnet?
- Valid hosts?

Practice Example #3: 255.255.128.0 (/17)

Network 172.16.0.0

Subnet	0.0	128.0
First Host	0.1	128.1
Last Host	127.254	255.254
Broadcast	127.255	255.255

Practice Example #4: 255.255.240.0 (/20) Network 172.16.0.0

- Subnets? $2^4 = 16$.
- Hosts? $2^{12} - 2 = 4094$.
- Valid subnets? $256 - 240 = 0, 16, 32, 48, \text{etc.}, \text{up to } 240$.
- Broadcast address for each subnet?
- Valid hosts?

Practice Example #4: 255.255.240.0 (/20)

Network 172.16.0.0

Subnet	0.0	16.0		240.0
First Host	0.1	16.1		240.1
Last Host	15.254	31.254		255.254
Broadcast	15.255	31.255		255.255